TV Home 2.0 — Product Requirements Document

Oct 28, 2026 · @Aman Kumawat

Summary and goals

TV Home 2.0 turns an old Xubuntu PC into a living-room box that looks and works like Google TV, and is set up and used with a remote alone. Version 1 proved the idea: streaming sites run full screen in Chrome, F12 returns home, and settings pages exist. It falls short on design, typing, and at-a-glance status.

What v1 testing showed

- Joining Wi-Fi from the TV failed in practice: there is no on-screen keyboard and little feedback.
- Wi-Fi, Bluetooth and volume state are invisible until you open Settings.
- Leaving an app depended on F12 or Alt+F4, and the close logic broke once (Chrome rewrites its process titles).
- The TV-only session showed a black screen on the 2012 Intel graphics with the compositor off.
- Home is one row of tiles; Settings is a separate full page rather than a quick panel.

Goals for 2.0

1. Every daily task works with a remote only: open and close apps, join Wi-Fi, pair Bluetooth, change volume, power off. Zero keyboard or mouse after first boot.
2. Wi-Fi, Bluetooth, volume and the clock are always visible in the Home top bar.
3. Home appears within 3 s of the session starting; the Home key returns to Home within 1 s.
4. The user never ends on a dead black screen; a recovery path always exists.
5. The shell (UI + backend) uses at most 350 MB of RAM at idle on the 4 GB machine.

Users, hardware and constraints

The product targets one specific old PC and a family audience, so every design choice must fit 4 GB of RAM, 2012 graphics and a 1366×768 screen.

Users

- **Owner (Aman):** sets the box up, adds apps, fixes problems. Comfortable with a terminal but should not need one.
- **Family members:** non-technical, use only the remote. Must never see a desktop, a terminal or an error they cannot act on.

Target hardware and software

Item	Value
Motherboard	Intel DH61SA (H61, LGA 1155)
CPU and graphics	Pentium G620, 2.6 GHz, 2 cores; Intel HD Graphics (Sandy Bridge)
RAM	4 GB DDR3 (single stick)
Storage	WD Green 240 GB SATA SSD
Display	Samsung S19B150N, 1366×768, VGA only
Network	USB Wi-Fi adapter (Realtek RTL8192CU); no built-in Bluetooth
OS	Xubuntu 26.04 LTS: X11, XFCE, LightDM, NetworkManager, BlueZ, PipeWire
Browser	Google Chrome (Widevine L3)

Constraints

- **X11 only.** The session must run on X11 with xfwm4; keep its compositor on, because turning it off gave a black Chrome window on this GPU.
- **DRM caps quality.** Chrome on Linux plays Netflix at 720p at most; JioHotstar may refuse Linux. The product cannot fix this.
- **Dead CMOS battery.** The clock resets after power loss; the UI must detect a wrong clock and fix it via NTP.
- **No Bluetooth hardware yet.** Bluetooth screens must handle "no adapter" gracefully and light up when a USB dongle is added.
- **Must keep working offline.** Fonts, icons and the UI ship locally; only the streaming apps need the internet.

UI research: how Android TV and Google TV look

Google TV's home is calm and content-first: a slim top bar with status and settings on the right, rows of 16:9 tiles, and a quick-settings panel that slides in from the right. TV Home 2.0 copies this structure and its rules, not Google's branding.

Pattern	What Android TV / Google TV does	What TV Home 2.0 takes
Safe margins	Keep key elements inside a 5% margin: 48 dp left/right and 27 dp top/bottom on a 960×540 canvas (layouts)	68 px left/right and 38 px top/bottom at 1366×768
Grid	12 columns of 52 dp with 20 dp gutters (layouts)	12-column grid; rows of tiles align to it
Top bar	2025 redesign: tabs Home, Live, Apps; icons for search, screensaver and quick settings; time on the right edge (Android Headlines)	Tabs Home and Apps on the left; Wi-Fi, Bluetooth, volume, clock and settings on the right
Quick settings	Panel with day, date and time on top, then a 2×2 grid of tiles: Screensaver, Wi-Fi, Accessibility, Bluetooth (9to5Google)	Right-side panel: date and time, then tiles for Wi-Fi, Bluetooth, Sound, Display, Power
Overlay panels	Right overlay = actions independent of the screen behind it; center overlay = urgent prompt (layouts)	Quick settings = right overlay; confirmations = center overlay
Rows	Browse template: rows stacked vertically, up/down between rows, left/right within a row (layouts)	Home = Favourites row + All apps row
App tiles	16:9 banner, 320×180 px at xhdpi, logo inside a safe area, text included (icon guidelines)	16:9 app banners with the logo centred on the brand colour
Focus	Scale (1.025, 1.05 or 1.1×), glow, outline or colour change; one focused element at a time (focus system)	Tiles: 1.1× scale + white outline + glow. Buttons: fill turns white, text dark
Directions	Each direction (up, down, left, right) should have a clear purpose (layouts)	Up = top bar, down = next row, right from top bar = status icons, Back = one level up

What open-source TV launchers add

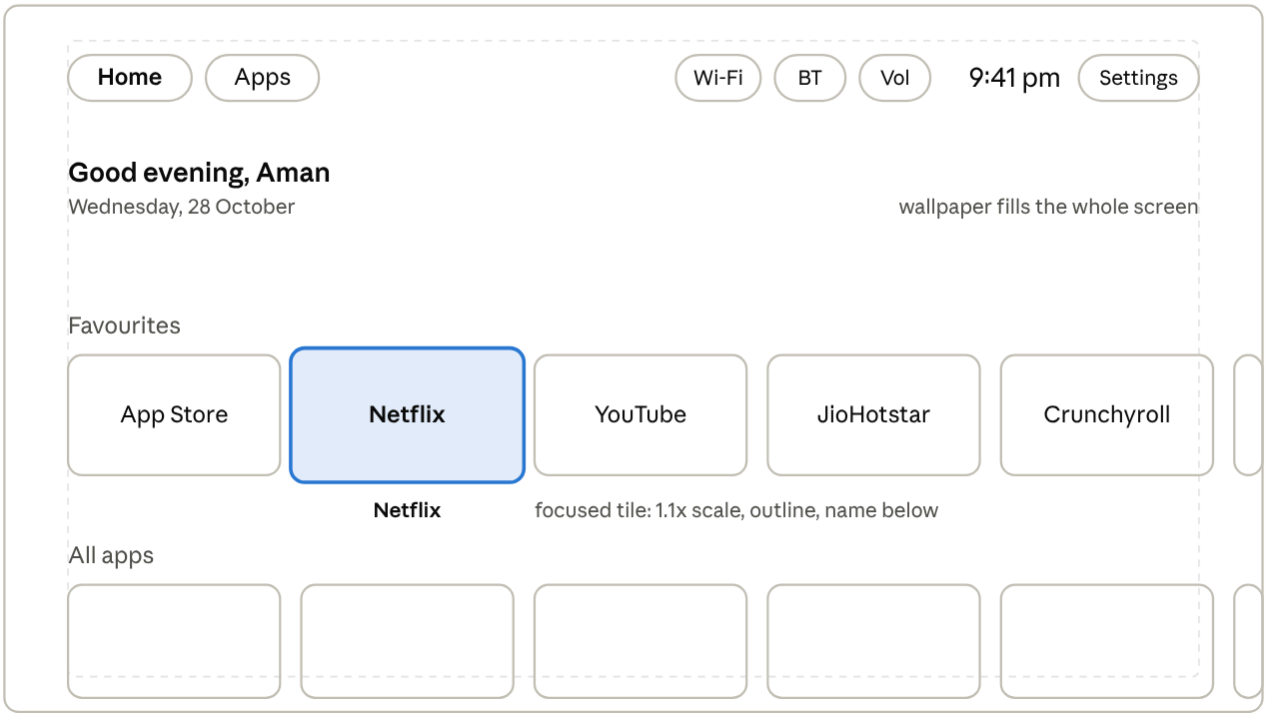
- HTLauncher (Windows): Bluetooth, Wi-Fi, Settings and Power in the top bar; Wi-Fi panel with signal bars and an on-screen keyboard.
- tvbox (Raspberry Pi): on-screen keyboard for all text, long values typed on a phone via a QR code, double-tap Back = Home.
- Plasma Bigscreen: a Home overlay reachable from anywhere to go home, switch apps and close apps.

Information architecture

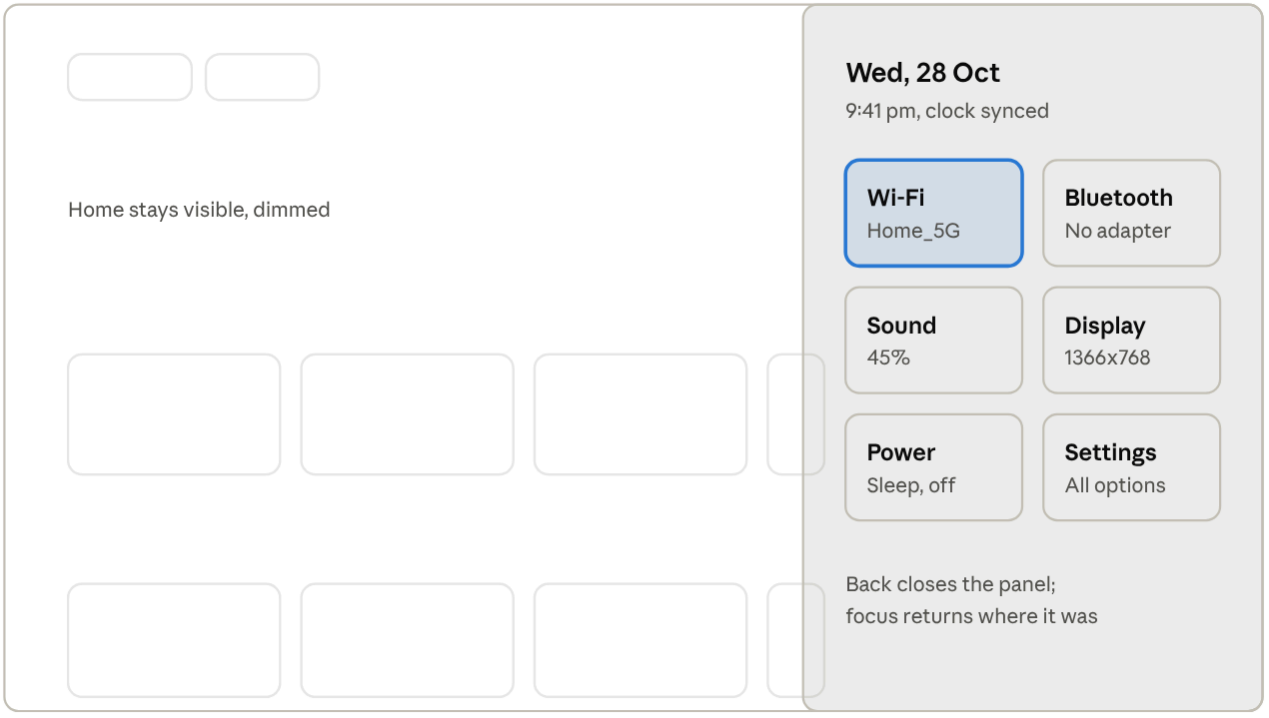
Home is the hub: a top bar with tabs on the left and live status on the right, rows of app banners below, and a quick-settings panel that slides over from the right.

Home keeps status in view; quick settings slide in from the right

1. Home at 16:9; the dashed line is the 5% safe area



2. Quick settings open: hold Home, or select Settings or any status icon



Home and quick settings wireframe at 16:9

The status icons double as shortcuts: selecting Wi-Fi, Bluetooth or volume opens quick settings with that tile focused.

Screen map

Screen	Opens from	What it holds
Home	Boot, the Home key, Back from any top-level screen	Top bar, Favourites row, All apps row
Apps tab	Top bar	Grid of every installed app, App Store tile first
App Store	App Store tile	Catalog by category, Add a website, install and remove
Quick settings (right panel)	Settings button, any status icon, hold Home	Date and time; tiles for Wi-Fi, Bluetooth, Sound, Display, Power, Settings
Settings pages	A quick-settings tile	Wi-Fi, Bluetooth, Sound, Display, Date and time, Wallpaper, About, Power: list on the left, detail on the right
Launch screen	Selecting an app	App logo and name until the app window appears
Running app	Launch screen	The website in Chrome kiosk; the Home key closes it
Confirm dialog (center)	Power actions, remove app, forget network	Title, one line, main button and Cancel
On-screen keyboard (bottom)	Any text field	Keys, Show password, Type on phone

Navigation rules: Back always goes up one level, Home always returns to Home, and focus returns to whatever opened a panel.

Visual design system

Dark, calm and content-first, with one warm accent; the focused item is always the loudest thing on screen.

Canvas and layout

- Design at 1920×1080 and scale by viewport height, so 1366×768 and future 1080p TVs both work. Set the root font size to 1.85vh and size everything in rem.
- Safe area: 5% on every side (68 × 38 px at 1366×768). Backgrounds and wallpapers bleed to the edges.

- 12-column grid. App banners use 2 columns each, so 5–6 fit per row with the next one peeking in.

Colour tokens

Token	Hex	Use
bg	#101418	Page background
surface	#1B2028	Panels, cards, keyboard
surface-raised	#262C36	Focused list rows, pressed buttons
text	#F1F3F5	Primary text
text-muted	#A9B1BC	Secondary text, row titles
accent	#FFB547	App Store tile, selected toggles, progress
good	#6FDCB8	Connected, on
warn	#F5C451	Weak signal, clock not synced
bad	#FF8A80	Errors, Shut down button

Typography (font: Lexend, bundled locally; fallback Ubuntu, sans-serif)

Role	Size at 1080p	Weight
Clock (top bar)	34 px	400
Screen title	44 px	700
Row title	26 px	500
Body, list rows	24 px	400
Labels, captions	20 px	500

Nothing smaller than 20 px at 1080p (about 14 px on the current screen).

Components

- **App banner:** 16:9, 12 px corner radius, logo centred on the app's brand colour (taken from the icon's edge pixels). The name appears under the banner only when focused.
- **Status icons:** Material Symbols (Apache 2.0), bundled as SVG: Wi-Fi with 0–4 bars and an off state, Bluetooth off / on / connected, volume off / low / high.
- **Buttons:** pill shape; unfocused = surface fill; focused = white fill with dark text.
- **List rows (settings):** full width, 64 px tall, label left, state right, 4 px gap.

Focus and motion

- Tiles: scale 1.1, 3 px white outline at 3 px offset, soft shadow, 150 ms ease-out.
- Rows and buttons: colour change only, no scale, so lists stay steady.
- The focused tile always scrolls fully into view, never under the safe margin.
- No looping animations on Home (they cost CPU on the Pentium); honour reduced-motion.

Functional requirements

There are 31 requirements; the 16 marked P0 make up the first release that fully replaces v1.

ID	Area	Requirement	Priority
H1	Home	Top bar: Home and Apps tabs on the left; Wi-Fi, Bluetooth and volume icons, clock and settings gear on the right	P0
H2	Home	Favourites row of 16:9 banners; move or remove an app by holding Enter	P0
H3	Home	Apps tab: grid of all installed apps plus the App Store tile	P1
H4	Home	Wallpaper with greeting and date behind the rows; local images only	P1
H5	Home	Status icons update within 3 s of a change (Wi-Fi drop, Bluetooth connect, mute)	P0
H6	Home	Short toasts for system events: Wi-Fi connected, headphones connected, low disk space	P1
H7	Home	Screensaver after 10 minutes idle on Home: clock over a dimmed wallpaper	P2
A1	Apps	Open a web app full screen in Chrome kiosk, with its own user agent when set (YouTube TV)	P0
A2	Apps	The Home key closes the open app and shows Home within 1 s	P0
A3	Apps	Launch screen shows the app logo until its window appears; clear error if nothing opens in 15 s	P1

ID	Area	Requirement	Priority
A4	Apps	App Store: curated catalog, add any website, install and remove, logos fetched automatically	P0
A5	Apps	Home overlay over a running app (hold Home): close app, volume, Wi-Fi status	P2
A6	Apps	Native Linux apps (Kodi, VLC) as tiles through their .desktop files	P2
S1	Settings	Quick settings panel from the gear or any status icon: date and time, then tiles for Wi-Fi, Bluetooth, Sound, Display, Power	P0
S2	Settings	Wi-Fi: networks with signal, lock and saved state; connect through the on-screen keyboard with show-password; forget; on/off; plain-language errors	P0
S3	Settings	Bluetooth: on/off, 10-second scan, pair + trust + connect, disconnect, forget, device-type icon, clear "no adapter" state	P0
S4	Settings	Sound: volume in 5% steps, mute, output picker (HDMI, speakers, Bluetooth)	P0
S5	Settings	Display: resolution list with a 15-second auto-revert confirmation; screen-off timer	P1
S6	Settings	Power: sleep, restart, shut down, go to desktop, each confirmed	P0
S7	Settings	Date and time: NTP sync status, timezone, warning when the clock is wrong	P1
S8	Settings	About: system, CPU, memory, storage, IP address, TV Home version	P1
S9	Settings	Wallpaper: choose from a folder; slideshow on/off	P2
I1	Input	On-screen keyboard for every text field: letters, numbers, symbols, Shift, Backspace, Done	P0
I2	Input	Type long text on a phone: scan a QR code on the TV, type on a local web page	P1
I3	Input	Remote key remapping with a learn mode	P2

ID	Area	Requirement	Priority
Y1	System	Boots straight to Home: no login screen, no desktop flash, no keyring prompt	P0
Y2	System	A watchdog restarts the shell; after 3 crashes in 5 minutes it falls back to the desktop	P0
Y3	System	Ctrl+Alt+D always reaches the desktop	P0
Y4	System	First-run setup: Wi-Fi, pair a remote, pick apps	P1
Y5	System	One-command installer and uninstaller; backup before any system file change	P0
Y6	System	Update TV Home from GitHub releases with one button	P2

Input: remote, Home and Back, on-screen keyboard

Everything is driven by six keys: four arrows, OK and Back, plus Home. Any remote that sends standard keyboard keys works; the recommended one is a ₹500–800 air-mouse remote with Home, Back and volume buttons.

Key map

Key (examples)	On Home and in TV Home screens	Inside a running app
Arrows	Move focus	Passed to the app
OK / Enter	Select	Passed to the app
Hold OK (600 ms)	Options for the focused tile: move, remove	Passed to the app
Back (Esc, Backspace, remote Back)	Close panel or go up one level	Passed to the app (the app's own back)
Home (XF86HomePage, F12, remote Home)	Scroll to the first tile	Close the app, show Home
Hold Home	Open quick settings	Open the Home overlay (P2)
Menu (Menu key, M)	Open quick settings	Passed to the app

Key (examples)	On Home and in TV Home screens	Inside a running app
Volume up/down/mute	Change system volume, show a volume toast	Same, globally
Ctrl+Alt+D	Exit to the desktop (recovery)	Same

Global keys (Home, volume, Ctrl+Alt+D) are grabbed at the X11 level so they work over any app. Focus never gets lost: when a panel closes, focus returns to the element that opened it.

On-screen keyboard (I1)

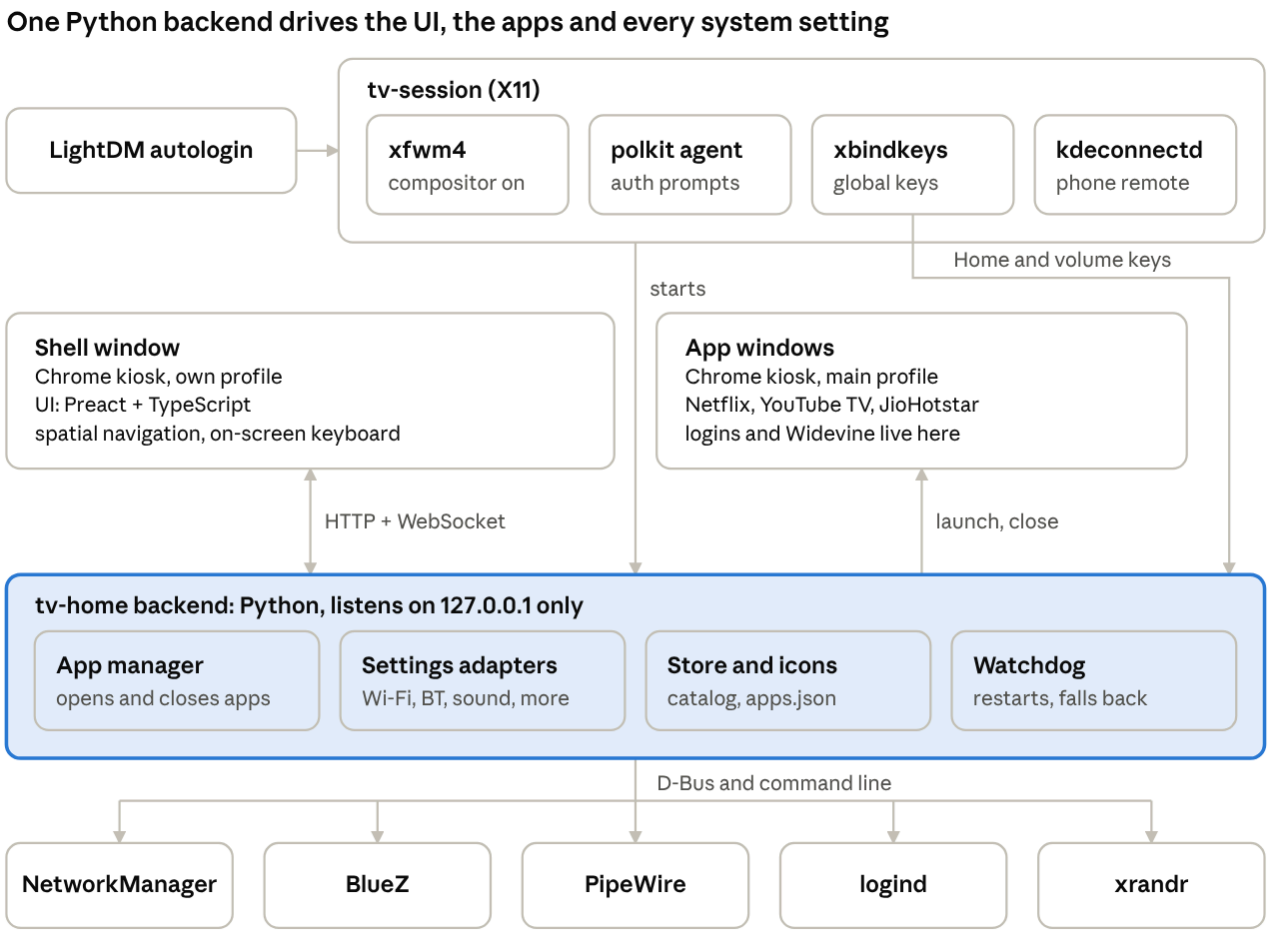
- Opens automatically when a text field is focused; slides up from the bottom and keeps the field visible above it.
- Layout: QWERTY in 4 rows, then a bottom row of 123/symbols, Shift, Space, Backspace, Done. Keys at least 56 px at 1080p.
- Arrows wrap around rows; Backspace on the remote deletes a character, holding it closes the keyboard.
- Password fields get a Show/Hide key; the last typed character shows for 1 s.
- Done submits the form (for Wi-Fi: Connect).

Phone typing (I2)

- Any text field offers "Type on phone": the TV shows a QR code and a 4-digit code.
- The phone opens a local page (same Wi-Fi), enters the code, types, presses Send; the text lands in the field.
- The local page runs only while the QR is on screen and closes after one use.

Technical architecture

A local Python backend owns all system access; the UI is a web app in its own Chrome kiosk window, and streaming apps run in separate Chrome kiosk windows. This reuses the Chrome and Widevine already installed and keeps RAM lower than Electron would.



TV Home 2.0 architecture on Xubuntu

The session script starts the window manager, helpers and backend; the backend opens the shell window, launches and closes apps, and talks to system services.

Stack

Layer	Choice	Why
Session	LightDM autologin into a custom X11 session script	No login screen or desktop flash; proven in v1
Window manager	xfwm4 with compositor on	Off gave a black screen on this GPU
Backend	Python 3, FastAPI + uvicorn in a venv	Fast to build, async, easy WebSocket push
System access	D-Bus via dbus-fast for NetworkManager, BlueZ and logind; wpctl for PipeWire; xrandr for display	D-Bus gives live change events and avoids parsing CLI output

Layer	Choice	Why
Frontend	Preact + TypeScript + Vite, fonts and icons bundled	Small bundle, works offline
Focus engine	Norigin Spatial Navigation (open source) or an in-house engine based on edge gaps	Arrow-key focus across rows, panels and the keyboard
Shell window	Google Chrome, kiosk, separate profile, extensions off, `--password-store=basic`	Already installed; no extra runtime
Global keys	xbindkeys posting to the backend	Works over any full-screen app on X11
Packaging	One install script, systemd user units, uninstall script	Easy to reinstall or remove

Key interfaces

- `GET /api/state` → apps, settings, status snapshot. `WS /api/events` → pushes Wi-Fi, Bluetooth, volume and clock changes.
- `POST /api/apps/{id}/launch`, `POST /api/home` (close the open app), `POST /api/apps` (install), `DELETE /api/apps/{id}`.
- `GET /api/wifi`, `POST /api/wifi/connect`, `POST /api/wifi/forget`; the same shape for `/api/bluetooth`, `/api/sound`, `/api/display`, `/api/power`.
- Data: `~/.config/tv-home/apps.json` and `settings.json`, written atomically; icons in `~/.config/tv-home/icons/`.

Non-functional requirements

The box must feel instant on a 2012 dual-core and must always leave the user somewhere they can act.

Performance budgets

Measure	Target
Power button to Home	40 s or less (SSD)
Session start to Home visible	3 s or less
Home key to Home visible (app open)	1 s or less
Focus move response	50 ms or less, no dropped frames

Measure	Target
Shell RAM at idle (UI window + backend)	350 MB or less
Shell CPU at idle on Home	3% or less
Status refresh after a Wi-Fi, Bluetooth or volume change	3 s or less

Reliability and recovery

- A watchdog restarts the shell if it exits or stops answering its health check for 10 s.
- 3 crashes within 5 minutes → start the full XFCE desktop instead, with a notice explaining why.
- Ctrl+Alt+D always ends TV Home and opens the desktop.
- Every change to a system file (LightDM, sessions, autostart) is backed up first and undone by the uninstaller.
- App data (apps, order, wallpaper, settings) lives in one JSON file, written atomically.

Security and privacy

- The backend listens on 127.0.0.1 only. The phone-typing page opens on the LAN only while its QR code is showing, behind a one-time 4-digit code.
- No root daemon. System actions go through NetworkManager, BlueZ and logind over D-Bus with polkit; the session starts a polkit agent.
- Wi-Fi passwords are handed to NetworkManager and never written to TV Home logs or files.
- App windows use the main Chrome profile; the shell uses its own profile, so streaming logins never mix with the UI.

Quality

- Works fully offline except for the streaming apps and icon downloads.
- Every error message says what happened and what to press next.

Build plan for Claude Code

Build in 8 milestones; each ends with something testable on a laptop in mock mode, then on the real PC. M0–M6 deliver all P0 requirements.

1. **M0 Foundation.** Repo layout, design tokens, focus engine, backend skeleton with a mock mode that fakes Wi-Fi, Bluetooth and sound.
   - Done when: `npm run dev` + backend in mock mode show an empty Home at 1366×768 and 1920×1080.

2. **M1 Home.** Top bar with tabs, status icons and clock; Favourites and All apps rows; banners with brand colours; wallpaper.
   - Done when: Playwright screenshots match the wireframe and every element is reachable with arrows only.
3. **M2 Apps.** Launch web apps in Chrome kiosk, launch screen, Home key closes apps, App Store with add-a-website.
   - Done when: Netflix, YouTube TV and a custom site open and close 10 times in a row on the real PC.
4. **M3 Quick settings + Sound + Power.** Right panel, volume, mute, output picker, sleep, restart, shut down.
   - Done when: volume keys and the panel stay in sync within 3 s.
5. **M4 Wi-Fi + on-screen keyboard.** Network list, connect, forget, errors; keyboard on every text field.
   - Done when: a WPA2 network is joined with the remote only, and a wrong password shows a clear error.
6. **M5 Session and recovery.** TV session script, LightDM autologin, xbindkeys, watchdog, desktop fallback, installer and uninstaller.
   - Done when: 10 cold boots reach Home with no desktop flash; killing the shell 3 times opens the desktop.
7. **M6 Bluetooth.** Scan, pair, connect, forget, no-adapter state.
   - Done when: a Bluetooth speaker pairs and plays from Netflix (with a USB dongle).
8. **M7 Polish.** Display settings with auto-revert, date and time, toasts, screensaver, phone typing, first-run setup.

Lessons from v1 to put in CLAUDE.md

- Chrome rewrites its process titles: identify app processes through `/proc/<pid>/exe` and match the whole command line as one string.
- Keep the xfwm4 compositor on; with it off, Chrome showed a black window on this GPU.
- Launch Chrome with `--password-store=basic`, or a keyring prompt appears at boot.
- `nmcli -t` escapes colons in SSIDs as `\:`; parse with an escape-aware splitter (or use D-Bus).
- The machine's clock can be hours off after power loss; check NTP sync before trusting TLS errors.

Testing approach

- Each system adapter (NetworkManager, BlueZ, PipeWire, logind, display) has a fake twin with recorded outputs, used in unit tests and mock mode.

- Playwright drives the UI with the keyboard only and saves screenshots at 1366×768 for review.
- A short manual checklist runs on the real PC before each release: boot, Wi-Fi join, app open and close, volume, shut down.

Out of scope, risks and open questions

Out of scope for 2.0

- Receiving casts (Chromecast, Miracast, AirPlay) and screen mirroring.
- Android apps (Waydroid, Bliss OS) and anything that needs Widevine L1, 1080p Netflix or 4K/HDR.
- User profiles, voice control, live TV and content recommendations ("Continue watching" rows).
- HDMI-CEC TV remotes (the current monitor is VGA only).

Risks

Risk	Impact	Mitigation
JioHotstar web may refuse Linux or need a Super/Premium plan	Key app missing	Test in M2; if blocked, document it and suggest a TV stick
Old Intel graphics glitch with some Chrome or compositor settings	Black or frozen screen	Keep the compositor on; watchdog + desktop fallback (Y2, Y3)
4 GB RAM with two Chrome instances (UI + app)	Stutter during playback	Shell profile with extensions off; one app at a time; measure in M2
An overlay above a full-screen kiosk window is hard on X11	Home overlay (A5) slips	Keep A5 at P2; use a small always-on-top window
Dead CMOS battery resets the clock	TLS errors, app logins fail	Clock check in S7; replace the CR2032 battery
Cheap remotes send odd key codes	Keys do nothing	Key remapping with learn mode (I3)

Open questions

- Name: keep "TV Home" or rename for a public GitHub release?
- License for publishing (MIT suggested)?
- Should the first release support more PCs than this one (other screen sizes, Wayland)?

- Which remote will be the reference for testing?

Sources

- Android TV design: Layouts
- Android TV design: Focus system
- Android TV design: TV app icon guidelines
- Android Headlines: New Google TV homescreen redesign reaches more users (Nov 2025)
- 9to5Google: Google TV homescreen redesign with new Quick Settings (Mar 2023)
- HTLauncher on GitHub
- tvbox on GitHub
- Plasma Bigscreen usage docs